B.Tech. ELECTRONICS ENGINEERING (VLSI Design &Technology)

COURSE PLAN: THEORY COURSE

Department :	**Electronics and Communication Engineering**			
Course Name & code :	**Digital Circuits & ECE2127**		**Core**	
Semester & branch :	**III & ECE**			
Name of the faculty :	**PRASHANT M PRABHU**			
No of contact hours/week:	**L**	**T**	**P**	**C**
	3	**0**	**0**	**3**

Course Outcomes (COs) to PO, PSO, BL Mapping

At the end of this course, the student should be able to:		**No. of Contact Hours**	**Marks**	**Program Outcomes (POs)**	**PSOs**	**BL *(Recommended)***
CO1	Explain the basic concepts of number systems, Boolean algebra, and various logical operations.	3	8	1,2	1	2,3
CO2	Apply logic minimization techniques like Karnaugh maps and Boolean algebra to design basic combinational circuits.	5	14	1,2,3,4	1	3
CO3	Construct combinational circuits, including multiplexers, decoders, and magnitude comparators, using Programmable Logic Devices (PLDs).	11	31	1,2,3,4	1	3
CO4	Analyze the operation of synchronous and asynchronous sequential circuits to understand their behavior and performance.	7	20	1,2,3,4	1	4
CO5	Differentiate between various types of flip-flops and sequential circuit models (Moore/Mealy) to optimize the design and implementation of state machines.	10	27	1,2,3,4	1	4
	Total	**36**	**100**			

Course Articulation Matrix

CO	Engineering knowledge	Problem analysis	Design/development of solutions	Investigations of complex problems	Modern tool usage	Engineer and society	Environment and sustainability	Ethics	Individual and team work	Communication	Project management and finance	Life-long learning				
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2										1	2			
CO2	3	3	3	2								1	3			
CO3	3	2	3	2								1	3			
CO4	3	3	1	3								1	3			
CO5	3	3	3	2								1	3			
Average Articulation Level	3.0	2.6	2.5	2.3								1	2.8			

ICT Tools used in delivery and assessment

Sl. No	Name of the ICT tool used	Details of how it is used
	LMS	For conduction of quizzes and assignments

Typical tools including LMS, Smart Boards, MS Teams, etc

Mapping of Course Outcomes (COs)/Course Learning Outcomes (CLOs) #

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes (POs)	Learning Outcomes (LOs)	BL *(Recommended)*
CLO1	Explain the basic concepts of number systems, Boolean algebra, and various logical operations.	3	8	1,2	**1**	2,3
CLO2	Apply logic minimization techniques like Karnaugh maps and Boolean algebra to design basic combinational circuits.	5	14	1,2,3	**1,2**	3
CLO3	Construct combinational circuits, including multiplexers, decoders, and magnitude comparators, using Programmable Logic Devices (PLDs).	11	31	1,2,3	**1,2**	3
CLO4	Analyze the operation of synchronous and asynchronous	7	20	1,2,3	**1,2**	4

	sequential circuits to understand their behavior and performance.					
CLO5	Differentiate between various types of flip-flops and sequential circuit models (Moore/Mealy) to optimize the design and implementation of state machines.	10	27	1,2,3	**1,2**	4
	Total	**36**	**100**			

***Applicable to IET Accredited Courses (modules) Only***

Delivery and Assessment Plan of LOs #

Learning Outcome (LO) mapped to the course		**Delivery and assessment Plan**
LO	**LO statement**	
1	Apply knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Some of the knowledge will be at the forefront of the particular subject of study	**Classroom teaching, Quizes, assignments and examination**
2	Analyse complex problems to reach substantiated conclusions using first principles of mathematics, statistics, natural science and engineering principles	**Classroom teaching, Quizes, assignments and examination**
3	Select and apply appropriate computational and analytical techniques to model complex problems, recognising the limitations of the techniques employed	**Classroom teaching, Quizes, assignments and examination**

***Applicable to IET Accredited Programs Only***

Assessment Plan (As communicated from o/o AD-A, in every odd semester)

IN – SEMESTER ASSESSMENTS									
Sl. No.	**Assessment Mode**		**Assessment Method**	****Time Duration**	****Marks**	**** Weightage**	**Typology of Questions *(Recommended)***	****Schedule**	****Topics Covered**
1	**MISAC**	**1**	**Quiz**	**20 minutes**		**Objective:** 5M 10 MCQs × ½ = 5 marks **Descriptive:** 10 M (2 Questions of 2 marks +2 Questions of 3 marks)	Bloom's taxonomy (B) level of the question should be L3 and above.		
		2	**Mid-Term Test**			10 MCQs × ½ = 5	Bloom's taxonomy (BT) level of the question should be L3 and above.		
		3	**Quiz**	**20 minutes**		1 Question × 5M = 5 marks (Minimum 5 questions to be given)	Bloom's taxonomy (BT) level of the question should be L3 and above.		
2	**FISAC**	**1**	**Quiz**	**20 minutes**	**5**	***	Bloom's taxonomy (BT) level of the question should be L3 and above.		
		2	**Assignment**	**	**5**	***	Bloom's taxonomy (BT) level of the		

							question should be L3 and above.		
END – SEMESTER ASSESSMENT									
1	**Regular/Make–Up Exam**			180 Mins	50	Answer all 5 full questions of 10 marks each. Each question can have 3 parts of 2/3/4/5/6 marks.	Bloom's taxonomy (BT) level of the question should be L3 and above.		

** ***Individual faculty will be entering the details***

*** ***Individual faculty shall identify the assessment method from FISAC Assessment method (Table 1 below) and fill in the details.***

NOTE: ***Information provided in the Table 1 is as per the In-semester assessment plan notified by Associate Director (Academics).***

Lesson Plan

L No	Topics	CO Addressed
0	*To address OBE philosophy towards NBA and IET accreditation followed by a discussion on Course Plan, Course Outcomes, CO-PO Mapping and Blooms Taxonomy, with prerequisites for the course, if any.*	
1	Review of Binary, Octal, Hexadecimal number systems and complement notations	CO1
2	Review of digital logic, Boolean algebra theorems and logical gates	CO1
3	Sum of products, Product of sums, Minterms and Maxterms expressions	CO1
4	Logic minimization using 2, 3, 4 & 5-variable K-maps	CO2
5	Minimization examples with & without don't care conditions to obtain minimal SOP and minimal POS expressions	CO2
6	Half-adder, Full-adder, Half-subtractor, Full-subtractor	CO2
7	Ripple-carry adder, Carry lookahead adder	CO2
8	Cascade of CLA adders, BCD adder	CO2
9	Magnitude Comparator, 1-bit, 2-bit and cascade connections	CO3
10	Decoders, 2:4 and 3:8 decoders using logic gates	CO3
11	4:16 decoder using 2:4 and 3:8 decoders	CO3
12	2-input and 3-input logic design using decoders, multiple output functions	CO3
13	Encoder, 8:3 priority encoder	CO3
14	Multiplexer, 16:1 mux using 4:1 and 8:1 multiplexers	CO3
15	Logic design using multiplexers	CO3
16	Logic design examples using mux and decoders	CO3
17	Programmable logic devices (PLDs), PROM	CO3
18	Programmable logic arrays (PLAs), 3-input logic design using PLAs	CO3
19	Programmable array logic (PAL), multiple output functions	CO3
20	Latches, SR latches using NOR/NAND, switch debouncer	CO4
21	Gated SR latch, D latch, propagation delay, setup time, hold time	CO4
22	Master-slave SR flip-flop, D flip-flop, JK flip-flop, 0's & 1's catching	CO4
23	Edge triggered D flip-flop, positive & negative edge triggering	CO4
24	Conversion from one type to another type flip-flop, design examples	CO4
25	Shift registers, SISO, SIPO, PISO & PIPO, universal shift register	CO4
26	Binary ripple counters, up/down, design examples	CO4
27	Synchronous counters, mode-N counters, shift-register counters	CO5
28	Design of synchronous counters using JK flip-flops	CO5
29	**Design of synchronous counters using D, T and SR flip-flops**	CO5
30	Self-correcting counters, design examples	CO5
31	Clocked synchronous sequential networks – Mealy & Moore models	CO5
32	Analysis of clocked synchronous sequential networks	CO5
33	Examples involving output expressions, transition equations, transition tables, excitation tables, state table, state diagram, etc.	CO5
34	**Serial binary adder as Mealy & Moore networks**	CO5
35	Binary sequence detector with & without overlaps	CO5

36	Asynchronous sequential circuit design – an overview	CO5

References:

Textbooks	• Donald D Givone, "Digital Principles and Design", Tata McGraw Hill, 2002. • William I. Fletcher, "An Engineering approach to Digital Design", Prentice Hall of India, 2009 • Samir Palnitkar, "Verilog HDL: A Guide to Digital Design and Synthesis", Prentice Hall 2003. • Stephen. Brown and Zvonko Vranesic, "Fundamentals of Digital Logic with Verilog Design", Tata McGraw Hill, 2013. • Charles Roth, Lizy K. John, Byeong Kil Lee, "Digital Systems Design Using Verilog", Cengage Learning, 2015
Self-Directed Learning: Hazards, Essential Hazards, Pulse mode sequential circuits, Design of Hazard free circuits	
Research Literature/ Case Studies	• •
NPTEL/Coursera/any MOOC-based material	• •

Submitted by: Prashant M Prabhu

(Signature of the faculty)

Date:

Approved by: Dr. Pallavi R Mane

(Signature of HoD)